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A novel damage index for damage identification using guided waves with application in laminated composites

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Abstract

In the current investigation, an innovative time-domain damage index is introduced for the first time which is based on local statistical features of the waveform. This damage index is called the 'normalized correlation moment' (NCM) and is composed of the n th moment of the cross-correlation of the baseline and comparison waves. The performance of this novel damage index is compared for some synthetic signals with that of an existing damage index based on the Pearson correlation coefficient (signal difference coefficient, SDC). The proposed damage index is shown to have significant advantages over the SDC, including sensitivity to the attenuation of the signal and lower sensitivity to the signal's noise level. Numerical simulations using Abaqus finite element (FE) software show that this novel damage index is not only capable of detecting the delamination type of damage, but also exhibits a good ability in the assessment of this type of damage in laminated composite structures. The NCM damage index is also validated using experimental data for identification of delamination in composites.

Keywords: damage index, damage identification, structural health monitoring, laminated composites, Lamb wave

(Some figures may appear in colour only in the online journal)

1. Introduction

Structural health monitoring (SHM) is an important area of engineering that is generally concerned with executing a paradigm of acquiring and processing structural dynamic data in such a way as to make informed decisions about the overall condition, or 'health,' of the structure [1]. Structural health monitoring, when implemented feasibly, results in considerable economic benefit in that it enables a paradigm shift from a fixed time-based maintenance to a condition-based maintenance strategy [2]. Most importantly, it may prevent structural failures which are often catastrophic.

Over the past few decades, as the knowledge of elastic waves has become better established, the potential of elastic waves to identify damage was brought to the attention of researchers and some novel game-changing damage

identification techniques were developed, taking advantage of the exceptional characteristics of elastic waves. Among the numerous traditional non-destructive evaluation approaches based on elastic waves, guided Lamb waves have probably been the approach most favored by researchers due to multiple advantages such as: cost-effectiveness, capability for inspecting large structures with fewer transducers, the ability to examine the entire cross-sectional area of the structure to detect surface as well as internal defects, the potential capability of classifying the types of damage using different wave modes, high sensitivity to both manufacturing defects and in-service damage, the possibility of inspecting coated or insulated structures such as underground pipelines, and use in automated damage detection and SHM techniques [3]. There are several studies describing the Lamb wave approach for identification of

damage in the literature [4–8]. For a more complete review of this area, refer to [9–11].

Ultrasonic Lamb waves have been shown [12] to be sensitive to various types of manufacturing defects and in-service damage in composite materials including voids, porosity, debonding, corrosion, cracking, hole, delamination, resin variation, broken fiber, fiber misalignment, resin crack, cure variation, inclusions, and moisture. Propagation of Lamb waves in anisotropic materials, however, is by far more complicated than in isotropic materials. The anisotropic nature of composite structures, especially in multilayered laminates, introduces many complex features and unique phenomena such as wave skewing, slowness, etc. Due to the various types of subtle damage that guided Lamb waves are sensitive to, and also due to their innate complexities, a large number of studies [13–17] (see [18] for a review) have been conducted on the propagation of Lamb waves in composite laminates and its application to SHM during the past decade along with the growing applications of advanced composites in various industries.

Actuators and sensors for damage identification using Lamb waves are often placed according to one of two configurations: (i) pitch-catch or (ii) pulse-echo. In either configuration, the transmitted or reflected waves need to be analyzed using certain damage identification algorithms to extract various characteristics (damage sensitive features) which contain essential information about the damage. Various damage identification algorithms have been used with guided waves in the literature which include time-of-flight (ToF) [19–23], damage index (DI), time-reversal techniques [24–27], migration techniques [28–32], Lamb wave tomography [15, 33, 34], probability-based diagnostic imaging [35, 36], phased-array beamforming [37, 38], and artificial intelligence techniques such as artificial neural networks [39–41], genetic algorithms [42–44], and the Bayesian interface [45]. ToF is the simplest and most straightforward feature of a Lamb wave signal that can be used for the localization of damage in one-dimensional and two-dimensional structures.

If a relationship can be established directly between the changes in some characteristics of the extracted waveform and some known damage parameters, this is called the damage index (DI). There are three types of damage indices: DIs in time domain, DIs in frequency domain and DIs in joint time/frequency domain. In this study, we are interested in time-domain damage indices. Various time-domain damage indices are used in the literature with different degrees of strength [46–51]. Some of them are as simple as peak-to-peak amplitude [52] and others are complicated enough to require nonlinear time-series analysis [51]. Time lag between two wave components in the signal, attenuation rate of the wave energy, and local statistical features are other characteristics of waves that form the basis of most damage indices. One promising time-domain DI is based on the Pearson correlation coefficient and is called the signal difference coefficient (SDC). The SDC is proven to be capable of early detection of fatigue damage in composite plates [53] or identification of

small simulated corrosion defects on an aircraft wing made of traditional isotropic material [50].

In the current investigation, an innovative time-domain damage index based on local statistical features of the wave form is introduced for the first time. This damage index is called the ‘normalized correlation moment’ (NCM) and is composed of the n th moment of the cross-correlation of the baseline and comparison waves. The performance of this novel DI in detecting waveform differences is compared with that of the SDC for synthetic signals and this novel damage index is shown to have significant advantages over the SDC, including sensitivity to attenuation of the signal and lower sensitivity to noise level. This novel damage index is used in some numerical simulations using Abaqus finite element software and is experimentally validated using data from testing for the identification of interlaminar delamination in composite laminates. This paper is organized as follows. In section 2, a concise background on the propagation of guided waves in general, and Lamb waves in particular, is provided for isotropic and anisotropic materials; in section 3, a brief summary of some existing time-domain damage indices is given; in section 4 the NCM damage index is introduced and the performance is compared with that of the existing SDC damage index in some limited trials using synthetic waveforms; in section 5, the capabilities of the new damage index are verified using an FE numerical simulation; and in section 6, the results are validated using data from experimental tests for identifying delamination in composite laminates.

2. Damage identification through guided waves

2.1. Waves in plates

The classical problem of the propagation of stress waves in solid media gives rise to the problem of guided and bulk waves. Although guided waves and bulk waves share the same set of partial differential governing equations, these two classes of wave are fundamentally different. Bulk waves travel in the bulk of material and away from boundaries. Guided waves, however, propagate along an elongated structure while guided by its boundaries. Mathematically, the introduction of boundary conditions to the problem of bulk waves leads to the guided waves problem. Thus, the solution to the governing equations in guided waves must satisfy some physical boundary conditions in addition to the governing equations. In contrast, there are no boundary conditions that need to be satisfied by the proposed solution in the case of bulk waves.

The problem of guided waves is much more difficult to solve analytically (than bulk waves) and in many cases the analytical solution to the problem of guided waves cannot be found. Some guided wave problems that have been solved are named after their investigators (e.g. Rayleigh, Lamb, and Stonely) [54]. Stonely waves are free waves that occur at an interface between two media and Rayleigh waves are free waves on the surface of semi-infinite solids. However, Lamb

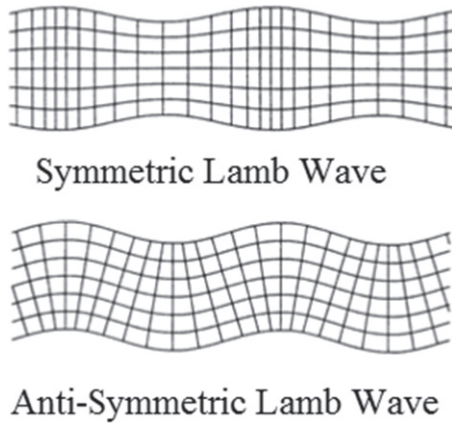


Figure 1. Symmetric and anti-symmetric Lamb waves.

waves are waves of plane strain that occur in a free plate with traction-free upper and lower boundaries. In this study, due to the geometry of the type of structures we are interested in, we will focus on guided Lamb waves.

Additionally, waves can be categorized according to the relative configuration of the direction of propagation and the direction of particle oscillations. In that sense, waves can be either longitudinal or shear. Longitudinal waves occur when the direction of particle oscillation is in the direction of wave propagation, while shear waves are formed when the direction of particle oscillation is perpendicular to the direction of wave propagation. Obviously, the oscillation particles in shear waves can be either perpendicular to the plane of wave propagation, called the vertical (or transverse) shear wave, or in the plane of wave propagation which is called the horizontal (or normal) shear wave.

2.2. Lamb waves

Lamb waves, which were discovered by Horace Lamb in 1917, are constituted from the superposition of a longitudinal wave and transverse (vertical) shear waves in thin plates and can exist in plate-like thin structures with parallel free boundaries.

A Lamb wave can be either symmetric or anti-symmetric as shown in figure 1. Symmetric modes (S) have identical strains at both the top and the bottom surface, while the strains at the top and bottom surface of anti-symmetric modes (A) have opposite signs. In a thin isotropic and homogeneous plate, the elasto-dynamics equation of waves, regardless of their modes, can be described (using Cartesian tensor index notation) as

$$\mu u_{i,jj} + (\lambda + \mu) u_{j,ji} + \rho f_i = \rho \ddot{u}_i \quad (1)$$

where u_i and f_i are the displacement and body forces in the direction x_i , and ρ , μ and λ are the density, shear modulus, and Lamé constant, respectively. The propagation of Lamb waves can be characterized by the phase velocity (c_p) and group velocity (c_g). Phase velocity and group velocity are functions of frequency and the considered wave mode. Phase velocity is the propagation speed of the wave phase of a particular frequency contained in the overall wave signal. For

a given wave and mode having the wave length λ , and the frequency f , the phase velocity is defined by

$$c_p = \lambda f = \frac{\omega}{k} \quad (2)$$

where k is the wave number ($k = \frac{2\pi}{\lambda}$) and ω is the circular frequency. Group velocity is associated with the propagation velocity of a group of waves with similar frequency. In other words, group velocity is the velocity with which the overall shape of the amplitudes of the wave (known as the modulation or envelop of the wave) travels and is defined by

$$c_g = \frac{d\omega}{dk} \quad (3)$$

Consider a Lamb wave in a plate structure excited at a specific central frequency, 300 kHz as an example. The power spectral density of the captured signal from the plate, if plotted in the frequency domain, will show a symmetric mode peak of 293 kHz and an anti-symmetric mode peak of 332 kHz, which are not identical to the original central frequency of 300 kHz [3]. This is due to the fact that the velocities of Lamb waves, as with most guided waves, are dependent on wave frequency and plate thickness. This phenomenon is called wave dispersion. Dispersion curves are used to describe and predict the relationship between frequency, phase/group velocity, and the thickness of various Lamb wave modes.

2.3. Waves in laminated composites

In a laminated composite plate, in addition to Lamb waves, another type of waves exists: horizontal shear waves, called Love waves. Love waves were first discovered by Edward H Love in 1911. Within each layer of a laminated composite plate, the particle displacements are governed by the Navier equation of motion as

$$\mu^n \nabla^2 u^n + (\lambda^n + \mu^n) \nabla (\nabla \cdot u^n) = \rho^n \ddot{u}^n \quad (4)$$

where u^n is the displacement and μ^n , λ^n and ρ^n are the shear modulus, Lamé constant and density for the n th layer in the laminate, respectively.

The anisotropic properties of composite structures introduce directional dependence and add to the complexity of the propagation of Lamb waves. For instance, in contrast to isotropic plates in which Lamb waves travel with the same velocity omni-directionally and the wave front forms a circle, in non-isotropic materials such as laminated composites, the wave velocity is subjected to the direction of propagation. A 'slowness' profile is commonly developed in order to visualize this discrepancy in the propagation of Lamb waves in different directions. A slowness profile is formed by plotting a function of direction-dependent velocity, $\frac{1}{c_g(\theta)}$, where θ is the propagation direction with respect to 0° .

In addition, the dispersion properties of Lamb waves in multi-layered structures are subject to the propagation direction when materials are anisotropic. Dispersion relations for materials of monoclinic and higher symmetry have been derived by Nayfeh and Chimenti in [55].

Table 1. Time-domain damage indices used in the literature.

Time-Domain Damage Index	Description
Peak-to-peak amplitude [52]	$DI = (A_1 - A_2)$
Amplitude combined with ToF [57]	$DI = (A_1 - A_2) \times (T_{F1} - T_{F2})$
Root mean square [58]	$DI_{RMS} = \sqrt{(1/T) \int [f(t)]^2 dt}$
Root mean square deviation [46]	$DI_{RMSD} = \sqrt{\int [f_2(t) - f_1(t)]^2 dt / \int [f_1(t)]^2 dt}$
Variance	$DI_{Variance} = (1/T) \int (f(t) - \bar{f}(t))^2 dt$
DI based on signal energy [47]	$DI = \left(\int f_1(t) ^2 dt \right) / \left(\int f_2(t) ^2 dt \right)$
DI based on signal energy [48]	$DI = \left(\int f_1(t) - f_2(t) ^2 dt \right) / \left(\int f_2(t) ^2 dt \right)$
DI based on signal energy [49]	$DI = \left(\int f_1(t) ^2 dt - \int f_2(t) ^2 dt \right) / \left(\int f_2(t) ^2 dt \right)$
DI based on Pearson correlation coefficient [56]	$DI = 1 - \left \sqrt{\int (f_1(t) - \bar{f}_1)(f_2(t) - \bar{f}_2) dt} / \sqrt{\int (f_1(t) - \bar{f}_1)^2 dt \int (f_2(t) - \bar{f}_2)^2 dt} \right $
DI based on time-reversal	$DI = 1 - \sqrt{\left(\int I(t) \cdot V(t) dt \right)^2 / \left(\int I(t)^2 dt \right) \left(\int V(t)^2 dt \right)}$ where $I(t)$ and $V(t)$ are the input and reconstructed Lamb wave signals after the time reversal.
DI's based on attenuation [59]	$DI = \ln A_b / A_d$
DI based on nonlinear time series analysis [51]	$DI = (S_i^u - S_i^d) / S_i^d$ where $S_i^u = \sum_{j=1}^{N_p-1} \sqrt{(w_{i,j+1}^u - w_{i,j}^u)^2 + (\dot{w}_{i,j+1}^u - \dot{w}_{i,j}^u)^2}$ and $(w_{i,j}^u, \dot{w}_{i,j}^u)$ denotes the j th point in the Poincare map

3. Feature extraction using time-domain damage indices

The aim of wave-based damage identification techniques is to detect changes between the incident and the captured (reflected or transmitted) waveforms occurring as a result of damage and also to determine how the extent of those changes can be linked to the damage parameters of the structure. To this end, some appropriate identification algorithms are used to seek certain characteristics of the waveforms which are associated with those changes and can be used to calibrate the difference between incident and captured waves. Those characteristics of the waveforms are called damage-sensitive features, or simply *features*, and this process of applying identification algorithms is called *feature extraction*. There are two main approaches for feature extraction. In the first approach, some pre-established physical models of the structure are used to extract features of the waves. This approach is called model-based feature extraction. In the second approach, features are extracted directly from signals without applying any deterministic model. This approach is called signal-based, or data-driven, feature extraction. If a relationship can be established directly between the changes in the features of the extracted waveform and some known damage parameters, this feature can serve as an indicator of damage and is called the damage index (DI). There are three types of damage indices: DIs in time domain, DIs in frequency domain and DIs in joint time/frequency domain. Various time-domain damage indices are used in the literature

along with a brief description as listed in table 1. Note that in this table, $f_1(t)$ and $f_2(t)$ refer to the time-domain baseline waveform and the comparison signal, respectively, while A_1 and A_2 are the peak-to-peak amplitude of the baseline and the comparison signal, T_{F1} and T_{F2} are the time-of-flight of the baseline and the comparison signal, and T is the duration of signal acquisition.

Various damage indices listed in table 1 have been used for damage identification in the literature with mixed levels of success. One of the most promising damage indices to date is a damage index based on the Pearson correlation coefficient first investigated by researchers at Penn State University [56]. The Pearson correlation coefficient is a statistical feature composed of the covariance of the baseline signal with that of the comparison signal divided by the standard deviation product of the two. Unity minus the absolute value of the Pearson correlation coefficient is called the signal difference coefficient (SDC) and is used as a damage index.

Consider $f_1(t)$ and $f_2(t)$ to be the baseline and the comparison waveforms used for the identification of damage. The signal difference coefficient between the two waveforms $f_1(t)$ and $f_2(t)$ is

$$SDC = 1 - \left| \frac{Cov(f_1, f_2)}{\sigma_{f_1} \sigma_{f_2}} \right| \quad (5)$$

where σ_f is the standard deviation of the signal and $Cov(f_1, f_2)$ represents the covariance of $f_1(t)$ and $f_2(t)$.

Therefore, the SDC can also be written as

$$SDC = 1 - \left| \frac{\sqrt{\int (f_1(t) - \bar{f}_1)(f_2(t) - \bar{f}_2) dt}}{\sqrt{\int (f_1(t) - \bar{f}_1)^2 dt \int (f_2(t) - \bar{f}_2)^2 dt}} \right| \quad (6)$$

where $\bar{f}$ is the mean value of the signal. In essence, the SDC is a statistical comparison between the signal in the present state and the signal in the reference state. Although the relationship between the SDC and any particular material or structural property is not yet clear, it is an indication of the overall change in the structure.

The SDC has been proven in multiple studies to be capable of identifying damage in some challenging cases. In the SDC is applied for a wave-based identification of fatigue cracks in a laminated composite plate during a fatigue-to-failure test and was shown to be capable of early detection of fatigue damage. In [50], the SDC is applied in combination with some tomography algorithms for identifying corrosion defects on aircraft wings and proved to be capable of identifying even small simulated corrosion defects. Due to multiple reports of successful application of the SDC to challenging problems involving guided-wave damage identification techniques, it is compared to our innovative time-domain damage index described in the next section.

4. A novel time-domain damage index

In this section, a novel time-domain damage index is introduced which is based on cross-correlation of the baseline and the comparison waveforms collected from the structure. In signal processing, cross-correlation, $r_{xy}(\tau)$, of two real-valued signals, $x(t)$ and $y(t)$, is a measure of similarity between the two sequences and is defined as

$$r_{xy}(\tau) = \int_{t=-\infty}^{+\infty} x(t)y(t-\tau)dt \quad (7)$$

Performing the cross-correlation involves computing a series of shifted (sliding) dot products, where the parameter τ is typically referred to as the ‘lag’ parameter. The cross-correlation function is maximized when the two signals have similar frequency content and are in phase with one another. In telecommunication, cross-correlation is used in developing matched filters in radar systems for the detection of the Doppler distorted echo of a known transmitted waveform.

Here we need to use a damage index for detecting changes in a captured waveform, compared with the incident waveform where these changes might be of types other than frequency content and phase change. Wave attenuation, for instance, is an important factor that plays a significant role in the damage-induced changes in a guided wave and can serve as a potential damage indicator. Hence, when comparing the captured waveform with the baseline waveform, we need to find a way to account for the overall shape of the waveform envelope as well as the frequency content. One way to do this is to use the concept of moment. Moment, statistically speaking, is a quantitative measure of the shape of a

distribution. The distribution shape of the cross-correlation of two signals has some valuable information that needs to be preserved. Here we take advantage of the n th moment of cross-correlation which preserves this information by taking more of the quantitative shape of the cross-correlation distribution into account. Therefore, the time-domain damage index, ϵ , that we call the ‘normalized correlation moment’ (NCM) is introduced as

$$\epsilon = \frac{\int_{\tau=0}^{\tau=T} \tau^n |r_{xy}(\tau)| d\tau - \int_{\tau=0}^{\tau=T} \tau^n |r_{xx}(\tau)| d\tau}{\int_{\tau=0}^{\tau=T} \tau^n |r_{xx}(\tau)| d\tau} \quad (8)$$

where r_{xy} is the cross-correlation of the baseline and comparison signal, r_{xx} is the cross-correlation of the baseline signal with itself (autocorrelation), T is the duration of signal acquisition, and n is the order of the statistical moment. Note that if the two signals are identical, the value of the damage index will be zero indicating no damage. Also note that the moment order, n , is arbitrary and not limited to integers.

As an illustration, and in order to compare the performance of this novel DI with that of an existing DI, namely the SDC, the NCM and the SDC are calculated for some synthetic trial signals in the remainder of this section. Consider a tone-burst ultrasonic signal $x(t)$ which can be generated by windowing a sine wave as

$$x(t) = w A \sin(2\pi ft + \phi) \quad (9)$$

where f is the center frequency of the tone-burst, w is the window function, A is the amplitude, and ϕ is the phase. A second tone-burst signal $y(t)$ is also synthetically derived from the first signal each time by altering one of the parameters of the $x(t)$ signal.

In the first trial case, the sensitivity of the two damage indices to wave attenuation is compared by increasing the amplitude A_y of the comparison signal $y(t)$ from 1% to 100% of the baseline amplitude. The values of SDC and NCM damage indices are calculated as the amplitude of the comparison signal is changing. The baseline and the comparison tone-burst signals for $A_y = 0.1 A_x$ are shown in figure 2. Also in this figure are the values of both DIs depicted. As can be seen from this figure, the SDC is completely insensitive to wave attenuation while NCM closely monitors wave attenuation. As the amplitude of the comparison signal increases from 1% to that of the baseline signal, the value of NCM decreases from 1 (which indicates totally different waves) to zero (which indicates completely identical waves).

Wave attenuation, which manifests itself as the gradual reduction in magnitude of wave signals, is an important phenomenon in composite structures. For an experimentally-measured table of attenuation rates in some typical composite layouts, refer to [59]. Attenuation in Lamb waves generally increases with the existence of damage or inhomogeneities such as stiffeners or fasteners [60–62]. Thus wave attenuation is an important feature in guided-wave SHM [59, 63–65] and the sensitivity to attenuation is a major advantage of NCM over SDC as a damage index.

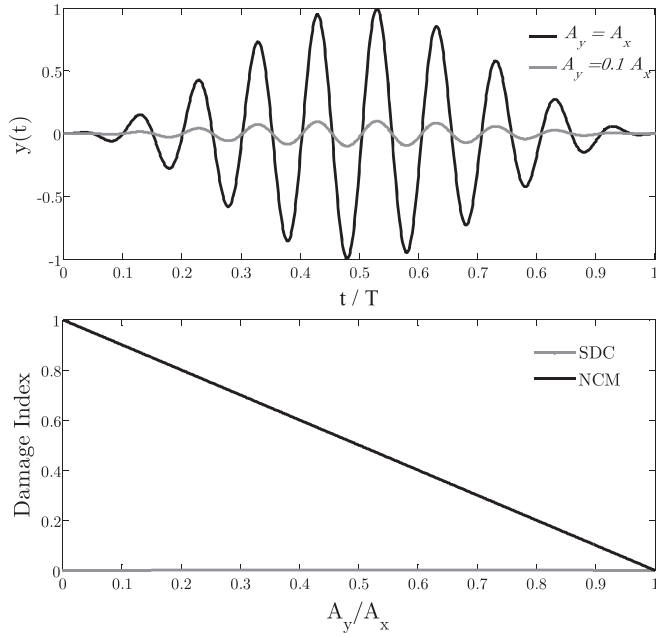


Figure 2. Sensitivity of damage indices to signal attenuation (the SDC index is a horizontal line at zero).

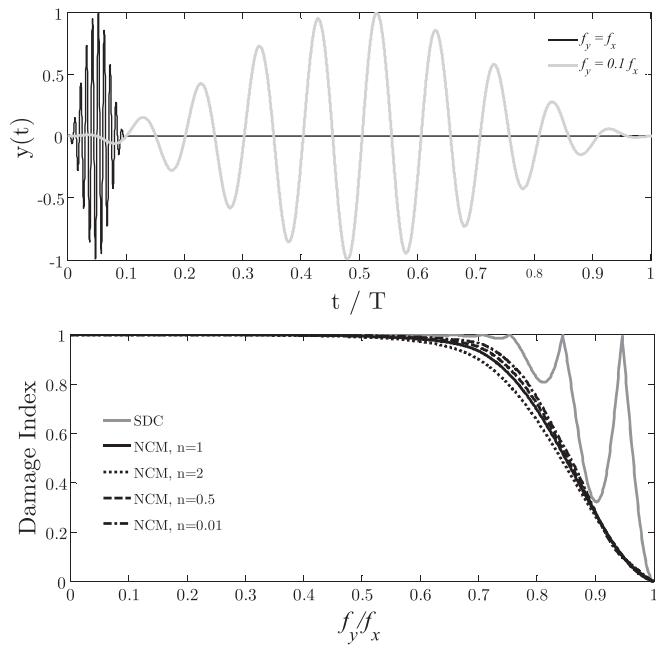


Figure 3. Sensitivity of damage indices to frequency difference.

In the second trial case, the sensitivity of the two DIs to frequency difference is compared by increasing the frequency of the comparison signal from 1% to 100% of the frequency of the baseline wave. The values of SDC and NCM damage indices are calculated as the frequency of the comparison signal is changing. The baseline and the comparison tone-burst signals for $f_y = 0.1 f_x$ are shown in figure 3 along with the variation of SDC and NCM damage indices as the frequency content of the comparison signal becomes a closer match. While NCM shows a monotonic sensitivity to frequency changes, SDC exhibits an oscillating sensitivity which

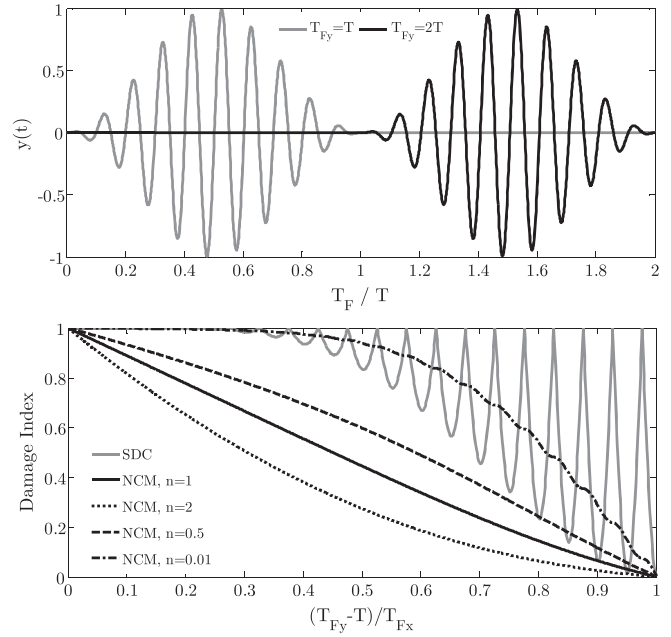


Figure 4. Sensitivity of damage indices to time-of-flight.

cannot be used as a reliable tool for drawing conclusions regarding the frequency content of two signals. Take further notice that as the moment order of the NCM decreases, the sensitivity of the NCM to frequency shows a slight increase.

As discussed earlier, ToF is a simple characteristic of Lamb waves and one of the most widely-used features from the SHM point of view. It has been utilized in developing damage indices for both the existence and localization of damage. Using ToF directly as a damage index, however, may not be the best choice since ToF as a damage index is too simple and does not provide any information about the severity of the damage *per se*. However, a DI based on local statistical features that can measure damage severity, coupled with ToF, could provide a more sophisticated damage index which has the distinguishing capability of damage location.

Next, in the third trial case we study the sensitivity of the two DI to ToF which can also be considered as wave velocity. The ToF in the baseline signal is considered to be equal to the duration of the signal ($T_{F_x} = T$) and that of the comparison signal is increased from $T_{F_y} = T$ to $T_{F_y} = 2T$, as shown in figure 4. The values of the two DI are compared as the ToF increases (figure 4). While NCM shows a monotonic sensitivity to variations in the ToF of the waves, the SDC exhibits an oscillating sensitivity. Therefore, the SDC cannot be used as a reliable tool for drawing conclusions regarding the ToF of the two compared waves. Since ToF is one of the most widely used features in guided-wave SHM, this is clearly a significant drawback for the SDC and a distinguishing advantage of the NCM over the SDC. Another point that is notable in the figure is that for $n = 1$, the values of the NCM index vary almost linearly with the ToF. A linear dependency of a damage index on ToF is valuable in wave-based SHM because it means this damage index might have the potential of being used for the purpose of determining the location of damage in addition to the existence and severity of damage.

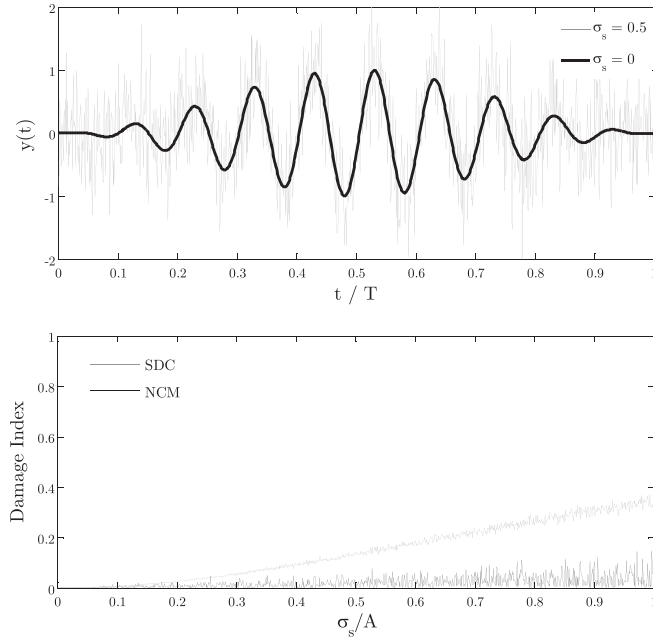


Figure 5. Sensitivity of damage indices to noise level ($\tilde{x}(t) = x(t)$).

Through establishing an inverse relationship between the NCM and ToF, the values of the NCM damage index can be inversely mapped to the relative location of the damage with respect to the sensor. A substantial consideration of this potential aspect of the NCM requires a detailed investigation which will be pursued in a separate paper.

In the next trial case, we study the effect of noise on the performance of these two DI by considering a comparison signal as

$$y(t) = \tilde{x}(t) + s(t) \quad (10)$$

where $s(t)$ is an additive noise constructed by creating a zero mean white noise time series with a standard deviation σ_s , equal to 0–50% of the peak-to-peak signal amplitude ($2A$). Two different cases are considered. In the first case, we assume $\tilde{x}(t) = x(t)$ and just noise is added (figure 5). However, in the second case, we consider $\tilde{x}(t) \neq x(t)$ (figure 6).

The comparison result for the case $\tilde{x}(t) = x(t)$ is depicted in figure 5 as the noise level, σ_s/A , is increased from 0 to 1, where σ_s is the standard deviation of the noise term $s(t)$ and noise level is defined as the ratio of σ_s to the signal amplitude A . While the value of SDC for these two identical waves decreases up to 37% as a result of noise, the NCM ($n = 0.01$) maintains the initial value of zero with less than 15% fluctuations. Note that since the two waveforms are basically identical, the values of the damage indices are ideally zero. For the case of $\tilde{x}(t) \neq x(t)$, we consider the frequency of $\tilde{x}(t)$ to be 10% less than the frequency of $x(t)$. The result depicted in figure 6 shows that while the SDC experiences a reduction in the mean value of the index and also an increase in the standard deviation of the index, the mean value of the NCM ($n = 0.01$) remains almost unchanged and only the standard deviation of its value increases. Standard deviation of the SDC is compared with that of the NCM for various orders of moment in table 2.

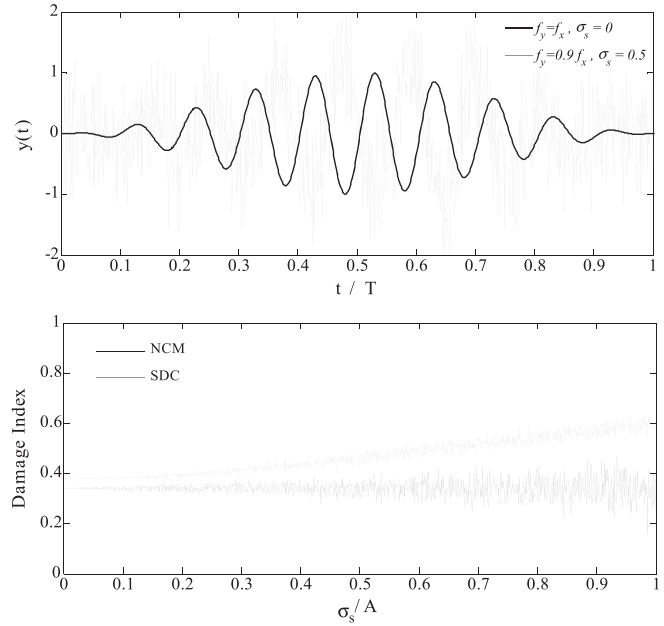


Figure 6. Sensitivity of damage indices to noise level ($\tilde{x}(t) \neq x(t)$).

Table 2. Standard deviation of the variation of indices as noise level (σ_s/A) increases.

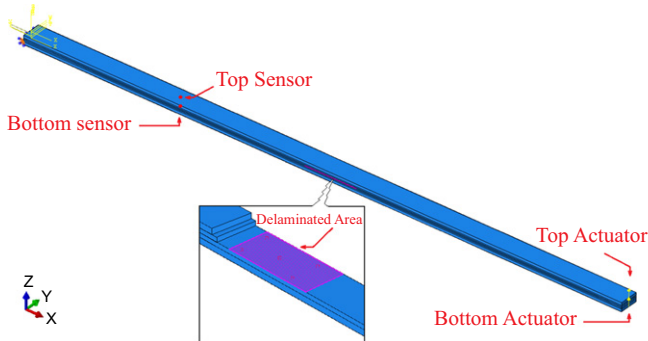
	SDC	NCM	
$x(t) = \tilde{x}(t)$	0.1109	$n = 0.01$	0.0242
		$n = 0.5$	0.0291
		$n = 1$	0.0365
		$n = 2$	0.0527
$x(t) \neq \tilde{x}(t) \ (f_{\tilde{x}} = 0.9 f_x)$	0.0701	$n = 0.01$	0.0299
		$n = 0.5$	0.0360
		$n = 1$	0.0436
		$n = 2$	0.0642

5. Simulation results

In order to demonstrate the application of the proposed damage index, NCM, in guided-wave identification of delamination in laminated composite structures, a numerical simulation using FE software was performed. The structure is a laminated composite cantilever beam having dimensions of $480 \times 15 \times 7$ mm made of Carbon/Epoxy IM7-8552 (see table 3 for material properties). The thickness of each lamina is 0.5 mm and they are oriented with a stacking sequence of $[0_4/90_6/0_4]_T$. The laminated cantilever beam is modeled in the Abaqus finite element package as shown in figure 7. The model is a 14-layer laminated cantilever beam modeled by eight-noded solid brick elements (C3d8) in Abaqus CAE. Damage is introduced as a rectangular area of delamination at the mid-span of the beam between layers 7 and 8 (i.e., at the mid-plane of the laminate) and is modeled using contact elements. Three different levels of damage are considered based on the area of the delaminated interface as listed in table 4. Layers with similar ply orientation are modeled as a single solid layer in Abaqus CAE using 8-node solid 3D elements. Therefore, the 14-layer laminated beam is modeled

Table 3. Material properties of Carbon/Epoxy IM7-8552.

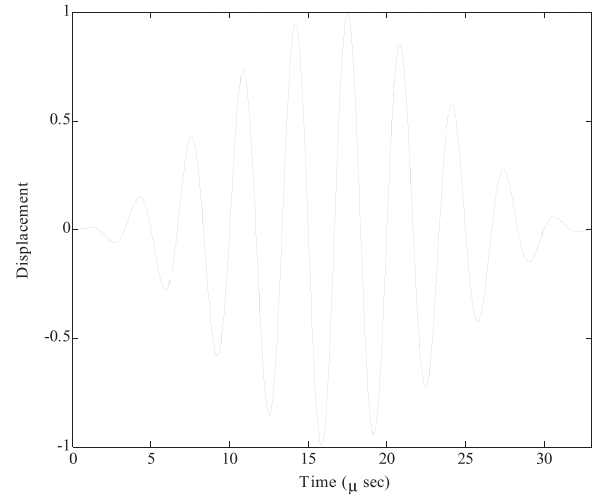
E_1 (GPa)	E_2 (GPa)	E_3 (GPa)	ν_{12}	ν_{13}	ν_{23}	G_{12} (GPa)	G_{13} (GPa)	G_{23} (GPa)	ρ (kg m^{-3})
171.420	9.080	9.080	0.32	0.32	0.5	5.290	5.290	3.973	1590

**Figure 7.** Laminated cantilever beam model.**Table 4.** Damage levels based on proportional area of delamination.

Damage Level	Delaminated Area (%)
Intact	0.0
Level I	6.25
Level II	12.5
Level III	25.0

with 4 layers of solid elements tied together at the interface nodes, so that the top four 0-degree layers are spanned by a single element through the thickness, the next three 90 degree layers are spanned by a single element through the thickness, the next three 90 layers are again spanned by a single element through the thickness, and the final four 0 degree layers are spanned by a single element through the thickness, as shown in figure 7. Nodes at the interfaces of delaminated areas are not tied together and instead they are ‘debonded’ by defining contact interaction between them to ensure that the no-penetration boundary condition is satisfied at the delaminated interface. The contact interaction considered in the FEA simulation is a frictionless surface-to-surface contact. In order to ensure the accuracy of the modeling, the spatial discretization was chosen to satisfy the rule of thumb condition of allocating at least 10 elements to span across the shortest wavelength of excitation waves. To this end, a dominant element size of 1.5 mm is chosen for each of the 4 solid layers. Hence, each of the 4 solid layers of the model is independently meshed with 3200 solid brick elements.

Since Lamb waves propagate at high velocities, the damage-scattered wave packet can easily become masked by wave packets reflected from boundaries [3]. Additionally, if the captured signal consists of several modes, which is usually the case, it is much more difficult to separate the effects of different modes. In order to best comprehend the interaction of a specific mode with simulated damage, it is of

**Figure 8.** Tone-burst excitation wave commonly used for damage identification.

great advantage if all the energy in the simulated incident wave is confined to a single mode [66]. It is not easy in practice, if not impossible, to separate modes in a captured waveform. However, it is much easier to separate symmetric modes from anti-symmetric modes by exciting symmetric or anti-symmetric modes only. A method called ‘pure mode excitation’ is used here. As mentioned before, the strain induced by symmetric Lamb wave modes at the top and bottom surface of a plate have identical signs but those of anti-symmetric modes have opposite signs. Therefore, we attempt to excite the ‘pure mode’ by applying the time-varying actuation profile of the desired mode (symmetric or anti-symmetric) at the top and bottom surface of the beam end. When the top and bottom actuating points have two in-phase equal-amplitude excitation signals, only the symmetric modes are excited. Conversely, when the excitation signals of the top and bottom actuating points are anti-phase equal-amplitude signals, only the anti-symmetric modes are excited.

A tone-burst signal having 10 cycles with a center frequency of 300 kHz windowed by a Hanning window is used as the excitation wave (figure 8). Simulated excitation is applied using two actuators (one on the top and one at the bottom of the free end of the beam) with special piezoelectric elements available in Abaqus. Two cases of equal-amplitude excitations are considered. In the first case, the tone-bursts from both actuators are in-phase and in the second case, the tone-bursts are anti-phase. The captured waves from the FEA cantilever laminated beam model for the intact and the damaged case (level 1) in 3 directions of U_x , U_y and U_z using in-phase and anti-phase excitation signals are depicted in figures 9(b) and (e), showing that, as expected, there is no

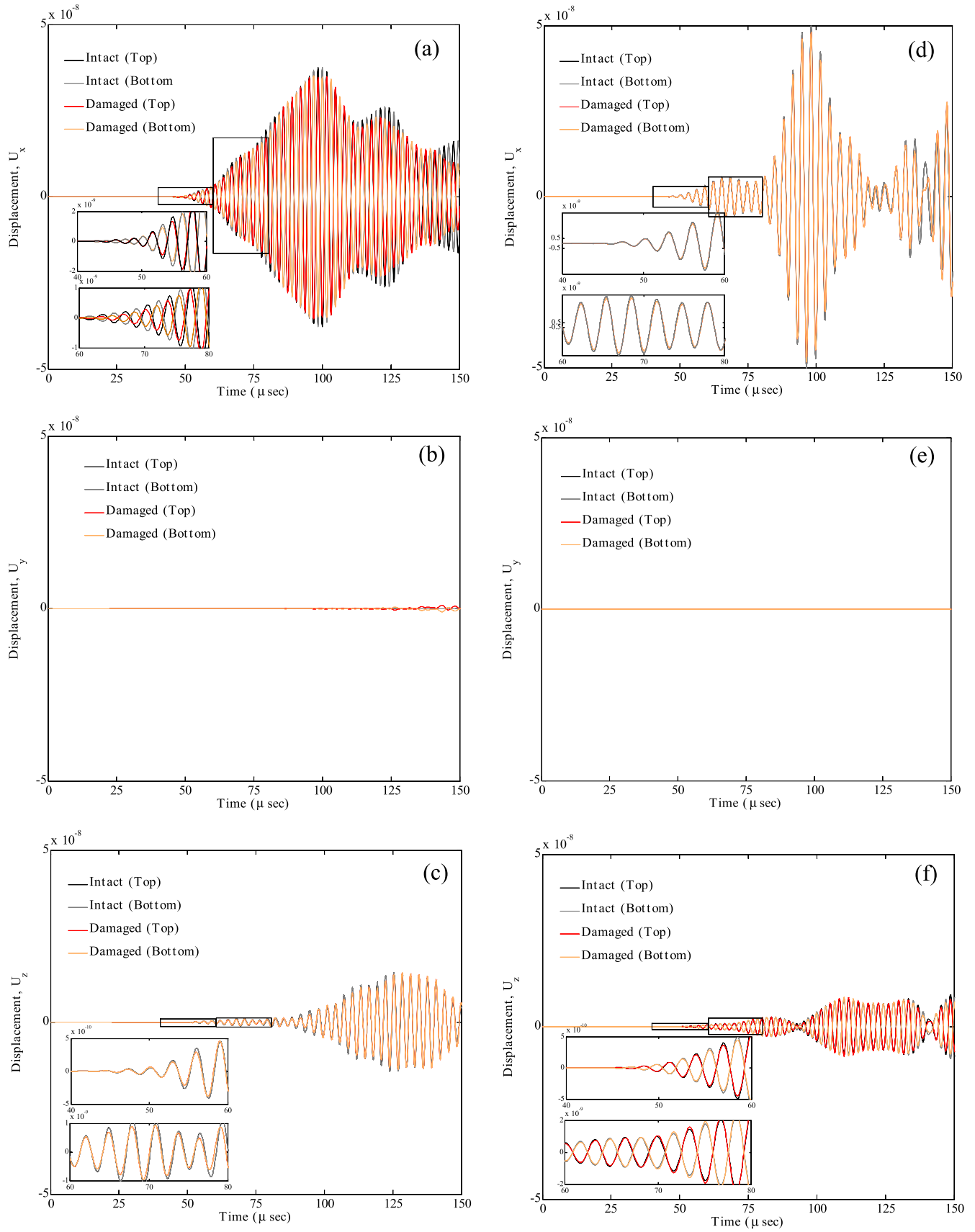


Figure 9. Propagation of a tone-burst wave along the cantilever laminated beam for intact case and damaged case (level 1) in 3 directions U_x ((a),(d)), U_y ((b),(e)), and U_z ((c),(f)) using equal-amplitude anti-phase actuators ((a), (b), (c)) and equal-amplitude in-phase actuators ((d), (e), (f)).

wave propagation in the y -direction (horizontal direction) in the captured wave. In other words, the captured wave is a Lamb wave and is only composed of longitudinal and vertical shear waves (and not horizontal shear waves). Note that there is no energy dissipation in FE simulation, unlike what occurs in reality. Therefore, the waves travel back and forth between boundaries repeatedly, without being dissipated and the captured Lamb wave is generally much more complex than the excitation due to multiple wave modes that coexist simultaneously and overlap each other.

As is clear from figures 9(a) and (c), the captured Lamb waves, when anti-phase equal-amplitude excitation signals are used, demonstrate purely anti-symmetric modes. However, captured waves shown in figures 9(d) and (f) which belong to the case of in-phase equal-amplitude excitation signals represent purely symmetric modes. Obviously, each of the captured symmetric and anti-symmetric waves might be composed of more than one mode. A power spectral density plot of the captured waves show that the frequency of the dominant symmetric mode is 286.5 kHz whereas the frequency of the dominant anti-symmetric mode is 327.4 kHz. The group velocity of the symmetric modes is 7683 m s^{-1} and that of the anti-symmetric modes is 8147 m s^{-1} .

When using time-domain damage indices for SHM, an important question that arises relates to the length of the captured signal to be considered in feature extraction. As discussed earlier, due to the high velocity of the waveforms, the damage-scattered wave packet can easily become masked by wave packets reflected from the boundaries. This accentuates the important role of the length of the captured wave that is considered for feature extraction. It is widely accepted that the first arrival packet of the captured wave contains the most valuable information for distinguishing a damaged structure from an intact one. Since the excitation wave is a tone-burst signal with 10 cycles, the length of the captured signal considered in this simulation for feature extraction is equal to the arrival time of the first 10 cycles of the captured wave. The NCM damage indices are calculated for the signal captured by the top and bottom sensors at each level of damage for the anti-symmetric modes and are listed in table 5. Similar damage indices for the symmetric modes are listed in table 6. As clear from these results for both symmetric and anti-symmetric captured wave modes, the NCM damage index was able to detect delamination in the beam model. Additionally, the values of the damage indices at different levels of damage show that the value of NCM is proportional to the delaminated area of the beam making it also capable of assessing the severity of delamination. An important point that can be concluded through comparing the values of NCM from anti-symmetric modes (table 5) with those from symmetric modes (table 6) is that anti-symmetric shear waves, as confirmed by other studies [67, 68], better identify delamination types of damage than symmetric shear waves, while there is not much difference in the ability of symmetric and anti-symmetric longitudinal waves to identify delamination types of damage.

6. Experimental validation of the results

6.1. Composite laminate fabrication

For testing purposes, carbon fiber reinforced composite laminates are fabricated using the vacuum assisted resin transfer molding (VARTM) process. Laminated composite samples are fabricated using IM7 sized unidirectional carbon fibers and SC-780 resin. The procedure begins with cutting 14 $10'' \times 10''$ layers of unidirectional carbon fiber, two layers of a porous release film (to allow easy removal of the laminate after curing), and one layer of a distribution mesh (to channel the resin flow throughout the fiber). A steel plate used as a mold is cleaned with acetone and a mold-release solution is applied to the mold. The fibers are placed on the mold in the desired orientation on top of the first layer of the release film. The second layer of the release film is then placed on top of the fiber stack followed by the distribution mesh. To enhance the resin flow in the inlet and outlet of the setup, two pieces of a spiral tube are placed along the inlet and outlet sides of the fiber stack. The outlet of the setup is connected to a vacuum pump via a vinyl tube after passing through a sealed resin trap while the inlet is connected to the resin reservoir. The entire setup is vacuum bagged and sealed using tacky tape sealant. More details regarding the VARTM process can be found in [69]. The VARTM setup is shown in figure 10.

Using the VARTM process, a 14-ply laminate $10'' \times 10''$ in dimension with a stacking sequence of $[0_4/90_6/0_4]_T$ is fabricated. In order to simulate delamination defects, two strips of $1'' \times 3''$ and $2'' \times 3''$ Teflon tape are inserted in between the 7th and 8th plies of fiber, that is, at the mid-plane of the laminate. The Teflon strips are placed in such a manner as to allow us to obtain three $10'' \times 1''$ coupons with 1 inch delamination at the midspan, and three $10'' \times 1''$ coupons with a 2-inch delamination at the midspan, and four intact baseline specimens, respectively. The Teflon tape is pulled out once the resin is cured to produce delamination within the composite panel. The square laminate panel is then machined into $1'' \times 10''$ coupons in order to have intact (baseline) and delaminated samples from the same laminate panel. This procedure reduces the probability of defects due to differences in the manufacturing process. Three $1'' \times 10''$ coupons with a 1-inch delamination, a 2-inch delamination, and one exhibiting no delamination (baseline) are used for damage identification tests (figure 11). As seen in figure 11, the delamination is barely visible to the eye.

6.2. Test setup and instrumentation

The three laminate coupons fabricated by the VARTM process are instrumented with piezoelectric bending disks. These 2-layer piezoelectric elements are 0.016 inch in thickness and 0.5 inch in diameter, and bow in and out like a drum head when actuated. Due to the higher sensitivity of anti-symmetric modes to delamination damages [67, 68], we attempt to excite the pure anti-symmetric modes by using two equal-amplitude anti-phase excitation signals at the top and bottom surface of the beam end. Therefore, three piezoelectric disks are used for

Table 5. Damage indices obtained at different damage levels from the captured waves by the top and bottom sensors using a tone-burst excitation wave ($f = 300$ kHz, $N_{\text{cycle}} = 10$) applied through anti-phase equal-amplitude actuations.

Damage Level	U_x (Longitudinal wave)								U_z (Shear wave)							
	Top Sensor				Bottom Sensor				Top Sensor				Bottom Sensor			
	$n=2$	$n=1$	$n=0.5$	$n=0.01$	$n=2$	$n=1$	$n=0.5$	$n=0.01$	$n=2$	$n=1$	$n=0.5$	$n=0.01$	$n=2$	$n=1$	$n=0.5$	$n=0.01$
Intact	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Level I	0.0710	0.0679	0.0650	0.0589	0.0711	0.0679	0.0650	0.0589	0.0861	0.0966	0.1043	0.1122	0.0860	0.0965	0.1042	0.1121
Level II	0.1411	0.1378	0.1348	0.1269	0.1411	0.1378	0.1348	0.1269	0.1719	0.1947	0.2116	0.2306	0.1718	0.1946	0.2115	0.2305
Level III	0.2651	0.2649	0.2661	0.2629	0.2652	0.2650	0.2662	0.2629	0.3206	0.3672	0.4017	0.4431	0.3205	0.3670	0.4016	0.4430

Table 6. Damage indices obtained at different damage levels from the captured waves by the top and bottom sensors using a tone-burst excitation wave ($f = 300$ kHz, $N_{\text{cycle}} = 10$) applied through in-phase equal-amplitude actuations.

Damage Level	U_x (Longitudinal wave)								U_z (Shear wave)							
	Top Sensor				Bottom Sensor				Top Sensor				Bottom Sensor			
	$n=2$	$n=1$	$n=0.5$	$n=0.01$	$n=2$	$n=1$	$n=0.5$	$n=0.01$	$n=2$	$n=1$	$n=0.5$	$n=0.01$	$n=2$	$n=1$	$n=0.5$	$n=0.01$
Intact	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Level I	0.0625	0.0644	0.0654	0.0664	0.0625	0.0644	0.0654	0.0664	0.0421	0.0203	0.0046	0.0067	0.0421	0.0203	0.0046	0.0067
Level II	0.1239	0.1262	0.1270	0.1268	0.1239	0.1262	0.1270	0.1268	0.0908	0.0499	0.0203	0.0121	0.0908	0.0499	0.0203	0.0121
Level III	0.2288	0.2311	0.2307	0.2274	0.2289	0.2311	0.2307	0.2275	0.1801	0.1118	0.0621	0.0277	0.1801	0.1118	0.0621	0.0277

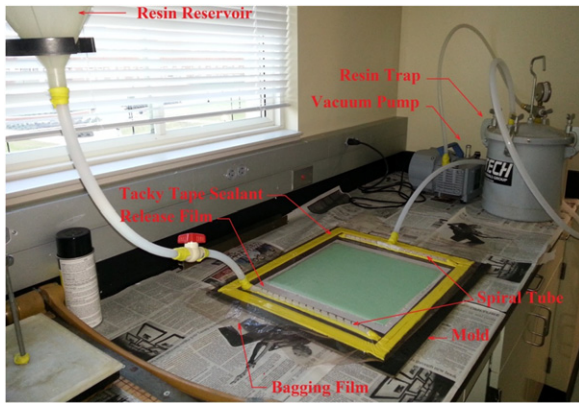


Figure 10. Vacuum assisted resin transfer molding (VARTM) setup.

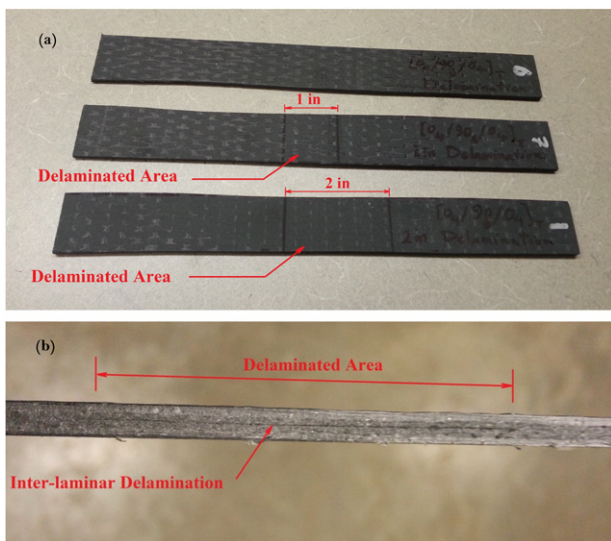


Figure 11. Composite laminate test specimens with interlaminar delamination: (a) test specimens with a 2-inch delamination, 1-inch delamination and with no delamination (baseline). (b) Through-the-thickness view of the delaminated test specimen.

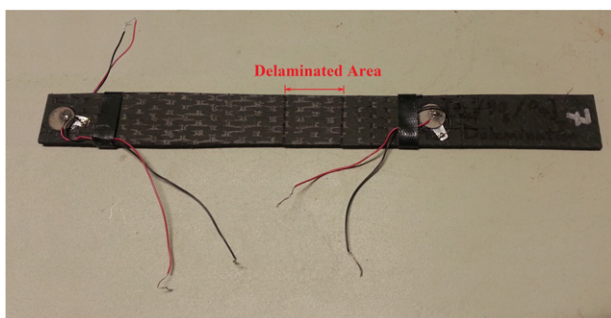


Figure 12. Laminate test specimen with 1-inch delamination instrumented with three piezoelectric disks.

each of the laminate coupons. Two are placed at the end of the specimen, at the top and the bottom, and serve as the actuators while the third one is placed at 3/4 of the length of the specimen and serves as the sensor. The 1-inch delaminated specimen instrumented with the piezoelectric sensor and actuators is shown in figure 12.

The piezoelectric disks are powered by an A-301 HS amplifier which supports an output bandwidth of up to 400 kHz. A NI PCI-6711 data acquisition (DAQ) card is used to generate the desired tone-burst excitation signal. The acquired voltage is measured by a NI PXIe-6250 DAQ card with 16-bit resolution and a high sampling rate which is capable of up to 1.25 mega samples per second. However, the DAQ card does not have a fast enough settling time at high scanning rates to be able to ensure 16-bit accuracy at maximum speed. Therefore, we use a sampling rate of 1 MHz for our measurements.

The excitation signal used in this experiment is a 10-cycle tone-burst signal. Although a high frequency Lamb wave (>200 kHz) is preferred for SHM purposes [70], to avoid interference with flexural modes of the structure, due to the limitations of our DAQ card, the highest excitation frequency is 50 kHz. Therefore, we use two tone-burst signals with center frequencies of 25 kHz and 50 kHz as the excitation signals.

6.3. Test results

A pitch-catch configuration is used for guided-wave experiments in this paper. The top and bottom actuators at the end of the test specimen are excited using equal-amplitude anti-phase tone-burst signals and the transmitted wave is measured by the third piezoelectric disk. The measured response for all three composite laminate test specimens is depicted in figure 13 for both excitation frequencies of 25 kHz and 50 kHz. All three responses are quite similar with the only difference being the amplitude of the response. Note further that due to the high velocity of Lamb waves and the relatively short distance between the PZT sensor and actuator there are no separate packets of waves for the incident waves and reflected waves in figure 13. In fact, the incident wave packet is overlaid by the wave packets reflected from the boundaries.

For ease of reference, the measured responses for each of the cases (intact, 1-inch delamination, and 2-inch delamination) are hereafter referred to as cases A, B, and C, respectively (table 7). For all combinations of these cases, the values of the novel damage index NCM for both excitation frequencies are calculated using equation (8). In order to compare the performance of the NCM with that of the SDC, the values of the SDC are also calculated for each case. The results are listed in table 8.

As expected from equation (8), the value of the NCM damage index for A-A comparison (intact response with itself) is zero while the SDC index for this case returns a small non-zero value, which is absurd. For the A-B case (intact response with 1-inch-delamination) the SDC index returns a meaningful value which indicates the existence of damage in the test specimen. However, since the difference in the captured waveform is mostly in the amplitude of the waves and it was previously shown in section 4 that the SDC index is completely insensitive to any changes in the amplitude of the waveform, the difference between the value of the SDC in the A-A case and that of the A-B case can probably be attributed to the slight phase difference between the captured wave from

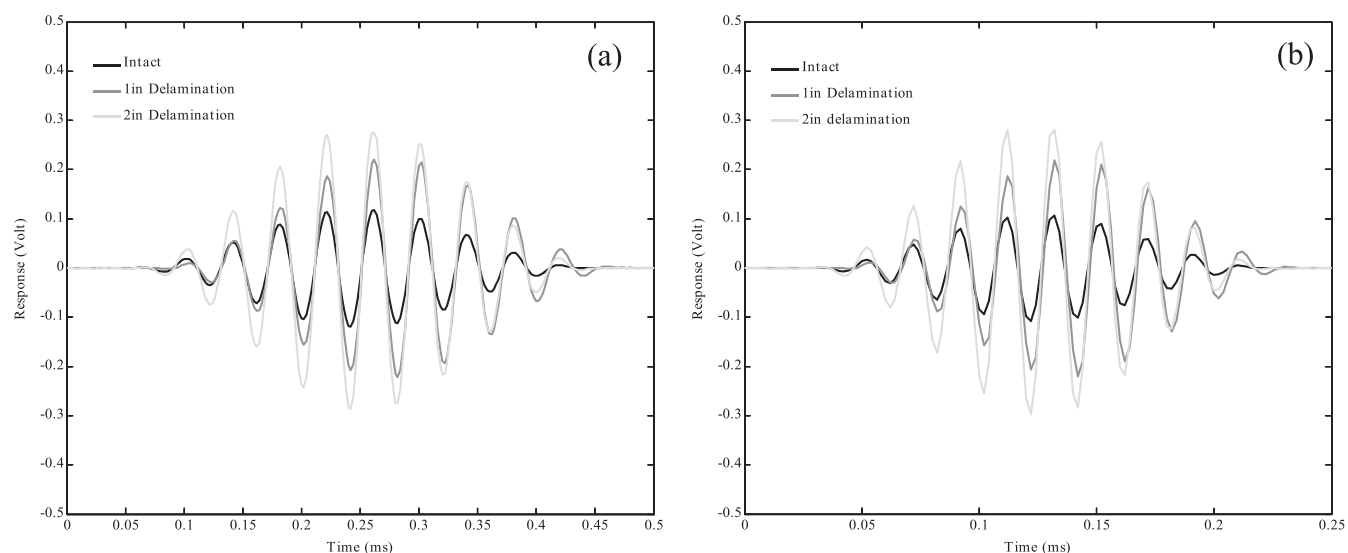


Figure 13. Transmitted wave measured from different test specimens using excitation signals: a) tone-burst with a center frequency of 25 kHz, b) tone-burst with a center frequency of 50 kHz.

Table 7. Damage cases based on delaminated length of the test specimen.

Damage Case	Delaminated Length (in)
A	0
B	1
C	2

Table 8. Damage indices for different comparison cases of the test specimens.

Frequency	Damage Index	Comparison Case		
		A-A	A-B	A-C
25 kHz	SDC	0.0040	0.0359	0.0053
	NCM ($n = 0.01$)	0	0.4942	1.3209
	NCM ($n = 0.5$)	0	0.4067	1.2930
	NCM ($n = 1$)	0	0.3299	1.2686
	NCM ($n = 2$)	0	0.2042	1.2273
50 kHz	SDC	0.0079	0.0679	0.0093
	NCM ($n = 0.01$)	0	0.5667	1.6831
	NCM ($n = 0.5$)	0	0.4906	1.6662
	NCM ($n = 1$)	0	0.4158	1.6492
	NCM ($n = 2$)	0	0.2887	1.6187

the intact and 1-inch delamination specimens. For the comparison case A-C, the phase difference is much smaller and therefore the value of the SDC index for this comparison case is only slightly larger than the value of SDC for the A-A case. The fact that the value of the SDC for the A-C case is smaller than that of the A-B case clearly shows that the SDC index in this experiment is not capable of assessing the severity of the damage and is only capable of identifying the existence of damage. The NCM index for both the A-B and the A-C cases returns meaningful values for both frequencies and all

moment orders which indicates the existence of damage. Additionally, the values of the NCM index for the A-C case is significantly larger than the A-B case for both frequencies and all moment orders. This means that as the severity of damage increases from a 1-inch delamination in the A-B case to a 2-inch delamination in the A-C case, the NCM damage index shows a larger value due to the greater amount of consequent changes appeared in the transmitted wave. Therefore, the experimental results clearly show that whereas the SDC index fails to assess the severity of interlaminar delamination damage in composite laminate test specimens, the novel NCM index is not only able to identify the barely-visible delamination damage but is also capable of assessing the severity of delamination defects in laminate test specimens.

7. Conclusions

An innovative time-domain damage index is introduced in this paper. This damage index, the ‘normalized correlation moment’ (NCM), is based on local statistical features of a waveform and is composed of the n th moment of the cross-correlation of the baseline and comparison waves. The performance of this novel damage index in detecting waveform differences is compared for various synthetic signals to an existing damage index based on the Pearson correlation coefficient (SDC) which is referred to as the signal difference coefficient (SDC). The novel NCM damage index is shown to have significant advantages over the SDC, including sensitivity to attenuation of the signal and decreased sensitivity to the signal’s noise level. This novel damage index is shown in numerical simulations using Abaqus finite element software to be not only sensitive to the existence of delamination, but also sensitive to the severity of delamination. This is generally a distinguishing advantage for a damage index. An experimental test for the identification of delamination in composite laminates confirmed that this novel damage index is a more

capable damage index than the SDC and provides a robust tool for both detecting delamination and assessing the severity of delamination in laminated composite structures.

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